Nowcasting

Instructions

Workshop 2 – Workshop on Useful Statistical Procedure for Nowcasting Part 2: Stationarity



IMF Institute for Capacity Development

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The dataset for this workshop is in "**stationarity.wf1**". This file contains three time series of daily (5-day per week) exchange rates: the exchange values of the Canadian dollar (**cadusd**), the British pound (**gbpusd**), and the Swiss franc (**chfusd)** against the US dollar, from January 4, 1971 to February 20, 2015 (44 years). In what follows, use only the data for the period January 1, 1976 to December 31, 2009 (you have additional observations in the workfile).

Question 1. Plot the ***(logarithm of)*** each of the resulting series on a single graph and "eyeball" them to decide whether they might be stationary or not.

Question 2. Compute the autocorrelation functions (ACFs) and partial ACFs (PACFs) of the **lchfusd** series. Do the ACFs decline slowly or quickly as the lag lengths increase? Do the PACFs show a pronounced drop-off after a certain lag length?

Question 3. Perform *non-augmented* Dickey-Fuller unit root tests, for models without and with a constant term. What is the null and alternative hypotheses used in these tests? What are the p values of the test statistics? Do you reject the null for either of the two tests?

Question 4. Perform *augmented* DF tests on this series, again without and with a constant term in the model. Use the AIC (Akaike Information Criterion) to let EViews decide the number of lags, p, of Δyt that should be used in the ADF regression. Is the value of p chosen by the AIC different from 0? Does your inference regarding the presence of a unit root depend on whether you use the ADF tests rather than the plain DF tests?

Question 5. Perform the KPPS test. The null hypothesis of stationarity is rejected: does this finding contradict the previous results?

Question 6. Forecasting with simple ARIMA models.

Part 1. Consider the autocorrelation functions (ACFs) and partial autocorrelation functions (PACFs) of d(lchfusd) since we found that lchfusd could be I(1).

Part 2. Estimate the model dlog(chfusd)=c for the period 1/1/1976 and 11/30/2009.

Part 3. Forecast **chfusd** for each day in December 2009. For this, use the sample 12/1/2009 - 12/31/2009. Plot the forecast series and the actual series over the forecast period

Part 4. Now let's compare it with the pure-random walk model, which assumes zero change from the end of November 2009 onwards (i.e., the optimal forecast for December 31, 2009 is actually the value of the exchange rate on November 30, 2009). The value of the exchange rate on November 30, 2009 was 1.005.